

Problem 0:

- Download Docker and
- Run docker without sudo

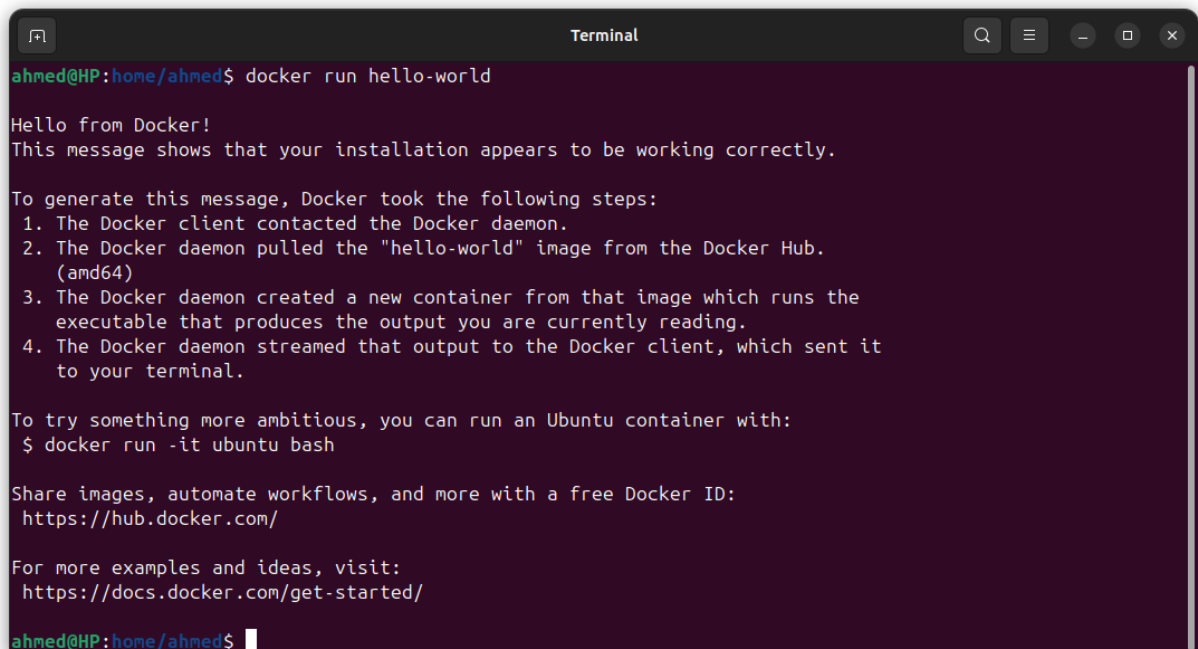
```
sudo apt-get update
sudo apt-get install ca-certificates curl gnupg lsb-release
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o
/etc/apt/keyrings/docker.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee
/etc/apt/sources.list.d/docker.list > /dev/null
sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-compose-plugin

sudo usermod -aG docker ahmed
newgrp docker
```

Problem 1:

- Run the container hello-world
- Check the container status
- Start the stopped container
- Remove the container
- Remove the image

```
docker run hello-world
docker ps -a
docker start -a 9b39f14eead5
docker rm 9b39f14eead5
docker rmi hello-world
```

A terminal window titled "Terminal" with a dark background. The prompt is "ahmed@HP:home/ahmed\$". The command "docker run hello-world" has been executed. The output is a multi-line message: "Hello from Docker!", "This message shows that your installation appears to be working correctly.", "To generate this message, Docker took the following steps:", followed by a numbered list of 4 steps describing the process from contacting the daemon to streaming output. Below the list, it says "To try something more ambitious, you can run an Ubuntu container with:" followed by the command "\$ docker run -it ubuntu bash". At the bottom, it says "Share images, automate workflows, and more with a free Docker ID:" followed by the URL "https://hub.docker.com/". Then "For more examples and ideas, visit:" followed by "https://docs.docker.com/get-started/". The prompt "ahmed@HP:home/ahmed\$" is visible at the bottom left of the terminal window.

```
ahmed@HP:home/ahmed$ docker run hello-world
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
ahmed@HP:home/ahmed$
```

Problem 2:

- Run container centos or ubuntu in an interactive mode
- Run the following command in the container “echo docker ”
- Open a bash shell in the container and touch a file named hello-docker
- Stop the container and remove it. Write your comment about the file hello-docker
- Remove all stopped containers

```
root@7d3431b12222: /
ahmed@HP:home/ahmed$ docker run -it ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
01d7766a2e4a: Pull complete
fd8cda969ed2: Download complete
Digest: sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9
Status: Downloaded newer image for ubuntu:latest
root@7d3431b12222:/# echo docker
docker
root@7d3431b12222:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@7d3431b12222:/# touch hello-docker
root@7d3431b12222:/# ls
bin boot dev etc hello-docker home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@7d3431b12222:/# exit
exit
ahmed@HP:home/ahmed$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
7d3431b12222   ubuntu    "/bin/bash"             5 minutes ago   Exited (0)    About a minute ago   vigilant_bose
ahmed@HP:home/ahmed$ docker rm 7d3431b12222
7d3431b12222
ahmed@HP:home/ahmed$ docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
ahmed@HP:home/ahmed$
```

Problem 3:

- Deploy a MySQL database called app-database. Use the mysql latest image, and use the -e flag to set MYSQL_ROOT_PASSWORD to P4sSw0rd0!. The container should run in the background.

`docker run --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0! -d mysql`

Problem 4:

- Run the image Nginx
- Add html static files to the container and make sure they are accessible
- Commit the container with image name IMAGE_NAME

```
root@7d3431b12222: /
ahmed@HP:home/ahmed$ docker run -d nginx
b8deda92e53379c654ca61391c8c651bcc3460ec82e3778f9416641585397d2c
ahmed@HP:home/ahmed$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS   NAMES
b8deda92e533   nginx     "/docker-entrypoint...."  12 seconds ago   Up 11 seconds   80/tcp   awesome_nash
ahmed@HP:home/ahmed$ docker exec -it b8deda92e533 bash
root@b8deda92e533:/# echo 'Ahmed Ehab Ahmed' > /usr/share/nginx/html/index.html
root@b8deda92e533:/# curl localhost
Ahmed Ehab Ahmed
root@b8deda92e533:/# exit
exit
ahmed@HP:home/ahmed$ docker commit b8deda92e533 IMAGE_NAME
invalid reference format: repository name (library/IMAGE_NAME) must be lowercase
ahmed@HP:home/ahmed$ docker commit b8deda92e533 image-name
sha256:1a7404cf15c2065d475daf0970eaacaf84b4b9ea6c6fff16da37408b7d354b19
ahmed@HP:home/ahmed$ docker image ls
IMAGE          ID              DISK USAGE  CONTENT SIZE  EXTRA
image-name:latest  1a7404cf15c2    237MB       62.9MB
mysql:latest      e5dc14f6e01c    1.27GB      283MB
nginx:latest       0236ee02dcbc    240MB       65.8MB      U
ubuntu:latest      d1e2e92c075e    119MB       31.7MB
ahmed@HP:home/ahmed$
```

Problem 5 (Bonus) :

- run a container with 2 volumes,
- vol-1 to change in html file, vol-2 change in nginx configuration make the port 8090
- remove the container, create new container with vol-1 for html files, and for vol-2 get the data from volume and then mount bind configuration
- make sure nginx is in running state

```
root@7d3431b12222: /
ahmed@HP:home/ahmed$ docker volume create vol-1
vol-1
ahmed@HP:home/ahmed$ docker volume create vol-2
vol-2
ahmed@HP:home/ahmed$ docker run -v vol-1:/usr/share/nginx/html -v vol2:/etc/nginx/conf.d -p 9090:8090 -d nginx
1faebf34300121e121a7886bf3bfd9b7ce1fd1d38bca30d5d606a369f1533545
ahmed@HP:home/ahmed$ docker exec -it 1faebf34300121 bash
root@1faebf343001:/# echo 'Ahmed Ehab Ahmed' > /usr/share/nginx/html/index.html
root@1faebf343001:/# vi /etc/nginx/conf.d/default.conf
bash: vi: command not found
root@1faebf343001:/#
```

apt-get update

apt-get install vim -y

vim /etc/nginx/conf.d/default.conf

```
root@7d3431b12222: /
server {
    listen      8090;
    listen  [::]:8090;
    server_name localhost;

    #access_log /var/log/nginx/host.access.log  main;
}
```

exit

docker rm -f 1faebf343001

mkdir -p ~/my-nginx-config

sudo cp -r /var/lib/docker/volumes/vol-2/_data/. ~/my-nginx-config/

```
root@7d3431b12222: /
ahmed@HP:home/ahmed$ docker run -v vol-1:/usr/share/nginx/html -v ~/my-nginx-config:/etc/nginx/conf.d -p 9090:8090 -d nginx
d2a9273939d49acb5bc989078f6e51240f49ec812d18736aabc111e1e658c618
ahmed@HP:home/ahmed$ curl localhost:9090
Ahmed Ehab Ahmed
ahmed@HP:home/ahmed$
```